

AIDEN J. BURGUEÑO

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SUMMARY

Bioengineering sophomore with strong critical thinking and problem solving skills, seeking an internship position to gain deeper knowledge in the field of bioengineering and the development of medical devices and technologies. Strong leadership and communication skills. Self-motivated and eager to learn. Effective when working independently and collaboratively.

EDUCATION

B.S., Bioengineering

Expected 6/2028

University of California, San Diego, La Jolla, CA

- GPA: 3.928
- Relevant Courses: Calculus for Science & Engineering Series, Differential Equations, Linear Algebra, Physics for Science & Engineering Series, General Chemistry I & II, Intro to Mechanical Design

Regents Diploma with Advanced Designation & Honors Math/Science

6/2024

Ward Melville High School, Setauket, NY

- GPA: 4.13/4.0
 - Relevant Courses: AP Calculus BC, AP Physics 1, AP Comp. Sci. A, AP Biology, Anatomy, Physiology, Mechatronics
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EXPERIENCE

RESEARCH EXPERIENCE

Cartilage Tissue Engineering Laboratory (PI: Dr. Robert Sah), UCSD, La Jolla, CA

Undergraduate Research Assistant

11/2024 – Present

- Contributed to study planning for osteoarthritis research using animal models to investigate cartilage degeneration.
- Assisted in experiments evaluating the biomechanical properties of the osteochondral interface.
- Conducting biomechanical testing on biological tissues to assess suitability as substitutes for damaged articular cartilage.

Summer Program Assistant

6/2025 – 8/2025

Led and mentored high school students through introductory research projects, guiding experimental design, testing procedures, and data analysis to support hands-on laboratory research.

LEADERSHIP EXPERIENCE

UCSD Biomedical Engineering Society (BMES), La Jolla, CA

Medical Innovation Conference (MIC) (formerly Translational Medicine Day) - Co-Chair

7/2025 – 2/2026

Led committee to plan and execute MIC; coordinated weekly meetings and outreach to biotech and medtech companies, startups, and professionals, delivering a conference with 12 presenters and 175 attendees.

Translational Medicine Day Committee - Industry Demonstrations Sub-Lead

10/2024 – 6/2025

Contributed to planning and execution of Translational Medicine Day; coordinated industry outreach and led the Industry Demonstrations team.

DESIGN AND MANUFACTURING EXPERIENCE

UCSD MAE 3, Intro to Mechanical Design

10/2025 – 12/2025

Course Robot Project - Team Member: Designed and built the drivetrain for a team competition robot, leading CAD and fabrication while using performance analysis to balance pushing strength, speed, and field obstacle clearance.

UCSD Rocket Propulsion Laboratory (RPL), La Jolla, CA

10/2024 – 6/2025

Project Daedalus - Team Member: Contributed to the design and fabrication of a single-use rocket, performing flight simulations in OpenRocket, designing components in SolidWorks, and assembling systems using soldering and 3D printing.

OTHER WORK EXPERIENCE

Physical Therapist Assistant, Tri-Fit Physical Therapy and Wellness, Setauket, NY

7/2023 – 9/2024

Deli Clerk, Wild by Nature Grocery Store, Setauket, NY

7/2024 – 9/2024

TECHNICAL AND PROFESSIONAL SKILLS

Certifications: Certified SolidWorks Associate, NY State Seal of Biliteracy (Spanish)

Design and CAD: SolidWorks, Fusion 360, AutoCAD

Manufacturing and Fabrication: Machining (lathe, drill press, band saw), Soldering, 3D printing, Laser Cutting

Programming: MATLAB, Java

Languages: English, Spanish